ANNEXURE - 2

GPS Renewables Pvt Ltd

Safety Requirements for Contractors

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Introduction

The following guidelines comprise the contractor's obligations towards safety practices at all GPS Renewables (GPSR) sites. GPSR is committed to a *Zero Harm* policy – zero fatalities, zero injuries and zero tolerance towards safety violations. To achieve this, GPSR will hold every contractor/subcontractor responsible and accountable for ensuring strict compliance with the established safety protocols and standard requirements, and all business-specific standards and procedures as may be provided by GPSR from time to time.

The contractor has to conform to all the requirements specified in the HSE SOP documents provided by GPSR. Some of the critical requirements (already part of the SOPs) are highlighted below for emphasis.

1. Permit to Work (PTW)

a. The contractor shall be familiar with permit requirements and be responsible for ensuring compliance with these requirements. No work shall be carried out without work permits authorized by GPSR.

b. There is an integrated work permit for all non-routine activities (except work pertaining to operation and maintenance). This integrated permit takes care of all related safety standards.

c. The work permit shall be valid for only one shift (time: 9 am to 6 pm. It can be renewed shift-wise: New permit for work under light (6pm to 6am) must be taken. However, GPSR will have the right to issue the work permit for a longer duration, based on the nature of work.

d. Except in the cases of routine jobs, where an integrated work permit is provided, no work permit shall be issued unless the contractor has a written job safety analysis (JSA) approved by GPSR, preferably in the local language prevalent at the site.

e. All non-routine work (except work pertaining to operation and maintenance) performed after 10pm shall require a separate 'night work permit' to be issued by GPSR and the same has to be procured and acknowledged by the contractor. To avoid doubt, it is clarified that when night work is required, an 'overlap' period shall be scheduled so that the day and night engineers can discuss the work and align themselves on the work to be performed under the night work permit. For Sundays and holidays, separate work permits shall be required which shall be authorized by GPSR's concerned line manager and/or safety officer.

f. The contractor has to give a list of authorised persons as 'Accepter' for the PTW. GPSR will authorise the issuer & receiver from their end.

2. Work at Height (W@H)

2.1. <u>Written plan:</u> The contractor shall have a written W@H plan – in addition to the relevant certificate(s) – certifying the right to W@H. Such a plan must ensure that:

a. Height works above 1.8 metres have safe access, egress, and a safe platform. Otherwise, workers must be secured through the use of a full body harness with a double lanyard, shock absorber and sufficient strengthened lifeline support.

b. The working platform shall have a hand rail, mid rail and toe board. A certified horizontal lifeline shall be

used, preferably an 8mm wire rope.

c. Working at more than 5 metres (e.g. transmission tower, truck covering etc.), or requiring different body posture even at lesser height shall require the use of the “Fall Arrestor” (for vertical movement).

d. Safety nets shall be installed to arrest all falling objects.

e. Step ladder/platform should be used if any personnel intend to work at height. Use of empty drums to climb up is banned. People taking tools for working at height shall have tool kits to facilitate three-point contact during access and egress.

f. GPSR’s floor-hole cover policy shall be complied with, and hole covers shall be fabricated as per the policy and design.

2.2. Competency checks:

a. Before starting work, the contractor must conduct formal competency checks of all personnel involved in designing fall prevention measures or fall protection measures. This will ensure that such personnel have the necessary training, experience and qualification to perform the assigned work.

b. The contractor shall also have a rescue plan in writing, which provides for, inter alia, the manner and mechanism in which all persons at the site, including people working at a height, are to be rescued.

3. Scaffolds

The contractor shall ensure compliance with the following standard practices for all scaffold-related activities and should conform to IS-3696 Safety codes for scaffold and ladders, IS-4014 Code of practice for Steel Tubular Scaffolding, IS - 2750 specification for steel scaffolding.

a. Every scaffold and its support shall be of good construction, made of suitable and sound material and strength.

b. Use of wood and bamboo scaffolds are banned.

c. Tube & coupler scaffold shall only be used with walkway planks. Cup-lock scaffolds shall only be used inside buildings on concrete surfaces.

d. All elevated structures/working platform areas should be guarded on all sides with railings and toe boards as described in the SOPs

e. Scaffolds shall be designed to support at least four times the anticipated weight of men and material. It shall have a standard inclined ladder/staircase for access and egress.

f. The landing platform shall be provided at every 6 meters of height. Extra support can be taken from existing nearby structures. Maintain a safe distance from power lines.

g. Fall protection when erecting, dismantling, modifying scaffolds shall be under the direction of a competent person.

h. The scaffold shall be designed, erected, and certified for use by competent persons.

i. Scaffolding that is ready for use shall be tagged with green. All other incomplete scaffolding shall be tagged with a red board and not be used until certified.

4. Portable Ladders

The contractor shall ensure that:

a. Only approved **aluminium ladders** are used at site which conforms to Indian Standard-3696 Safety codes for scaffold and ladders.
b. Metal ladders **shall not** be used by persons performing electric welding or working near energized electric lines or services.

c. Fabricated/site-made/painted ladders are not permitted.

d. A qualified person or inspection agency shall inspect all ladders as per the statutory requirements of GPSR.

e. Bends, dents, cracks, loose or missing rivets, disconnected braces, condition of steps and corrosion seriously weaken a ladder. Destroy or repair any defective ladders immediately in accordance with manufacturers' recommendations.

5. Barricades and Signs (conforms to IS-9457)

a. The contractor shall furnish, erect and dismantle all barricades as required for indicating and protecting from hazards such as excavated area, platform opening, crane swing area, fabrication yard, overhead work, work on road and other such locations.

b. Protective barricades shall be made of sheet metal or 40 NB pipe, painted with yellow and black stripes or red and white stripes, with caution signage on it. For excavated pits more than one metre depth, these barricades must be strong enough to resist a force of 100 kilograms.

c. For warning, indicative barricades can be made of fluorescent tape, cable and ropes with signs affixed.

d. No excavated pit shall be left unprotected or without signs indicating caution during the job, till backfilling is completed. Caution tape shall be used to highlight hazards to other personnel that may need to access the area.

e. Any barricade must be at least 1.5 meters from the edge of the excavation or opening. No overburden soil, or vehicle movement shall be allowed within 1.5 meters of an excavated pit.

f. Barricades, in areas with night traffic either by vehicles or people, must be lit with warning flashers (red or yellow) every 50 meters. There shall be caution boards and warning flags.

6. Confined Space (CS)

The contractor shall:

a. Confirm with the site engineer about working in confined spaces and follow site-specific confined space entry procedure

b. Follow safe entry procedure and obtain entry and work permits.

c. Provide on-the-job training to persons entering confined spaces and have a stand-by person and rescue team in the event of any incidents.

d. Ensure proper ventilation (forced or exhaust) and illumination using 24V power supply

e. Carry out checks for presence of toxic/flammable gases, monitor the oxygen content in confined spaces and ensure availability of communication media between stand-by confined spaces attendant and people inside confined spaces

f. Ensure availability of self-contained breathing apparatus or equivalent, and use of all relevant personal protective equipment as per JSA and PTW.

g. Ensure that all persons exit from the confined space upon completion of the work. To ensure such exit, head count of the men entering and exiting such confined spaces shall be performed and a record maintained in a register.

7. Excavation

Before an excavation begins, the contractor shall verify site-specific conditions such as traffic, vibrations, proximity of structures, soil, surface water and groundwater, chemical contamination of water, water table, overhead and underground utilities, weather condition etc. The contractor should ensure availability of barricading material before doing the excavation.

The following mandatory precautions must be taken & should conform to Indian Standard (IS) -3764(1992) for Excavation work -Code of Safety.

a. Providing high-visibility vests for people exposed to traffic.

b. Remove or stabilize all surface impediments, including removal of loose rock or soil that could fall and create hazards to persons

c. Erect either warning barricades (with depth of less than 1 metre) or rigid, protective barricades (with depth of more than 1 metre) to avoid leaving excavation hazards unprotected. However, a soil pile at least one point five (1.5) meter high can be used as a barricade on one side of the excavation.

d. Keep soil dirt and any material or equipment that may fall into an excavation at least 1.5 meter from the edge.

e. Excavated pits deeper than 1.5 meters will have a maximum vertical wall height of 1.5 meters with sloped edges at 45 degrees.

f. Protect, support, or remove underground installations (e.g., electrical ducts, water lines, sewer lines, or fire lines).

g. Prohibit personnel from working or passing under the loads of lifting or digging equipment inside the pit.

h. Provide support systems such as shoring, bracing, or underpinning to ensure the stability of adjoining buildings, walls or structures, particularly if depth is more than one metre. In such cases, confined space

entry permit shall be obtained and safe access/egress path shall be provided.

i. Ensure that no worker shall be allowed to work alone in a trench at any time.

8. Hot Work

8.1. Written work permit: The contractor shall obtain and adhere to a written work permit while carrying out any hot work involving welding, cutting, brazing, polishing or heating a metal drum, barrel, or tank. Sealed container, cylinder etc. must not be welded or gas cut in any case.

8.2. Training: All operators involved in hot work shall be trained on the hazard and risk control measures as per JSA and should be qualified. Flammable sources must be removed or protected from heat using fire retardant mat(s).

The following mandatory precautions must be taken and should conform to Indian S-818 Code of practice for safety and health requirements in electric and gas welding & cutting operations, IS-3016- Code of practice for **Fire precautions** in welding and cutting operations.

a. For gas cylinder safety, there shall be a flashback arrester both at the cylinder and torch. All precautions related to gas cylinder safety shall be adhered to. A two-stage pressure regulator shall be used with the cylinder.

b. GPSR's electrical engineer shall certify all welding machines for suitability and integrity.

c. Grinding/cutting wheels should conform with the BS EN-12413 and BS ISO-525 standards.

d. The contractor shall not use AC transformer-type welding machines. Only generator/rectifier type machines shall be used.

e. Electrical DB supply must be fitted with an ELCB/RCC, set at thirty milli-ampere (30mA).

f. Spatters generated from hot work must be arrested and the area shall be barricaded if hot work is performed at height.

g. The contractor shall ensure availability and use of recommended PPEs as outlined in the JSA and PTW

h. A trained fire watcher must be placed for all critical hot work. Job and machine grounding must be ensured.

i. In case of hot work in a confined space, an authorized person must carry out atmospheric tests.

j. The contractor shall provide fire extinguishers during hot work which shall be periodically inspected to check if it is fit to use.

9. Electrical installation

The contractor shall ensure that:

a. Any temporary power distribution system, including the source, shall be designed and approved by GPSR's electrical engineer.

b. All overhead cables should be over 2.5 meters high and on insulated or wooden support. No temporary cable shall be laid on the ground.

c. All electrical installations, distribution box and welding machines should be prominently marked for identifications with hazard signs, with the following information provided:
   i. Identification number
   ii. Name of the Subcontractor/contractor
   iii. Source of incoming/outgoing power
   iv. Danger sign in case above 440V DB.

d. Power supply points must be of adequate capacity, provided with proper switch and fuse arrangement and installed at strategic locations

e. Panel rooms/distribution box pathways are clear and free from any obstructions

f. While working near live electrical conductors, workers shall use arc flash suits of sufficient rating, as per JSA and PTW

g. Temporary electrical installations must meet the same electrical code requirements as permanent installations, i.e. proper earthing, glands and seals at electrical boxes and cable repairs equal to the original cable insulation.

h. Earth Leakage Circuit Breaker (ELCB) or Residual Current Circuit Breaker (RCCB): earth protection for personnel on the sites shall be provided by a thirty milli-ampere (30mA) ELCB/RCCB. It should not be replaced by MCB.

i. The Electricity Rules, 2005, and safety requirements specified under the applicable regulations issued by the Central Electricity Authority must be complied with, and the following conditions must be adhered to:

   i. Engage a wire-man and technician or engineer (license holder) to work on all electrical installations
   ii. Ensure only industrial type plug and sockets are used and proper earth is provided on electrical installations;
   iii. Electrical distribution boxes are installed on mountings at predetermined locations with its nomenclature
   iv. Rain protections are provided on electrical installations and DBs
   v. Only 3 core double insulated cables are used for any temporary wiring or portable/ hand held tools or lightings
   vi. Use authorized cable jointers/connectors to join the cables
   vii. Handles/body of portable electrical tools shall be of rubber or insulating material
   viii. Electrical systems shall be handled by a competent and authorized person for the voltage in line with applicable statutory requirements and conforms to Indian standard- 732 Electrical wiring installations.

10 Portable Tools and Equipment

The contractor shall ensure that:

a. Tools and equipment requiring formal periodic inspection are tagged, marked, labelled, tested, certified, and calibrated (as applicable) to indicate that the formal inspection has been performed.

b. All formal inspections have inspection records.

c. All certified, adequately insulated, fibre bodied, tested, checked and quality manufactured tools and tackles are used, including cutting, brazing, welding, cutting torch, rock breaker, driller, and lifting gear.

d. Any tools and equipment found defective or in need of inspection are tagged with a “defective” tag and returned to the appropriate person for repair or inspection. Tools or equipment must be removed from the site until repaired or inspected.

e. All portable grazing/grinding equipment has an adequate and valid wheel, insulation, guard cover, and good quality wire.

f. Cables for all power tools are without any cable joints and 2 Pin/3 Pin are used for power connection.

11. Hoisting/mobile lifting equipment

a. Contractor personnel are forbidden from riding on any hook, load, or headache ball.

b. Equipment operator certifications must be submitted to GPSR's site team.

c. Unauthorized lifts, including tandem lifts, are not permitted.

d. GPSR's site engineer must be notified prior to any critical lift (any lift exceeding 80% of the lifting device's design capacity or exceeding 2 tons).

e. All lifting equipment, tools, tackles, and devices shall be tested, certified, and maintained to manufacturer specifications.

f. For man lifts and other industrial lifts, the contractor shall procure a certificate from third party inspection agencies, as per statutory requirements, certifying successful testing of such lifts at the site prior to deployment.

12. Lifting Equipment

a. The contractor shall ensure all man and material lifting equipment, tools, and tackles are suitable for the job and approved by the site team.

b. The contractor shall use the GPSR site checklist for all cranes/heavy vehicle etc before hiring or bringing any such equipment to the site for use

c. Thereafter, the contractor shall use periodic and daily checklists to ensure the continued suitability of all equipment and tools.

d. All equipment and tools must have valid legal documents and maintenance records.

e. Operators must have necessary statutory certification, manufacturer's training record, refresher training every three years, and at least five years of experience.

f. All cranes must have a safe load indicator, overrun cut-off switch, and audio-visual alarms.

g. Safe working load, date of last testing, and due date for next testing must be marked on all lifting equipment.

h. First-generation hydra use is banned.

i. All material lift jobs must have a documented plan approved by the site team.

j. Critical lifts (over 2 tons or as agreed by the site team) must have a detailed material lift plan and are normally not permitted at night.

k. Cherry pickers/Pick & carry and any man-baskets used must be inspected by TPI before deployment.

13. Machine Guarding

The contractor shall be responsible for ensuring guarding of rotating machines adequately to prevent exposure of personnel.

14. LOTO

a. The contractor's supervisor shall ensure that lockout and tagout is performed according to the plan.

b. No personnel shall work beyond the facilities protected through lockout and danger tags.

c. All personnel involved directly or indirectly in the work associated with hazardous energy sources shall be involved in controlling these hazards, including developing/reviewing the lockout plan, placing locks and danger tags on isolation devices, or verifying the lock/tag on the equipment.

d. The contractor shall ensure proper isolation before allowing personnel to work on the equipment.

e. After ensuring proper lockout and tagout, the key to the lock shall be made available to the concerned supervisor or the engineer in-charge.

15. Concreting

a. Formwork shall be in line with the current approved drawing.

b. The contractor shall adhere to 'Bar Bending Schedule' and 'Concreting Plan' and shall submit and obtain approval for method statement, shuttering and staging plan including rebar spacing and chair design.

c. A competent person shall inspect formwork before pouring concrete.

d. All projected reinforcement bars shall be covered/bent when work is being carried out overhead.

e. Any timber formwork shall be inspected for cracks and excessive knots before use.

f. Tubular steel shall be inspected for rusting, dents, and damaged welds before it is erected.

g. All vertical supports shall be vertically braced with cross bar support.

h. Footings under the support shall be set on firm soil or other suitable materials which assures stability.

I. The prop shall be of full length and not joined together from multiple pieces.

j. During pouring of concrete there should be continuous inspection of centring wedges and adjustment screws.

k. Before removing the forms, the contractor shall ensure that the concrete has developed sufficient strength.

l. Formwork structure shall have sufficient catwalks and other secure access for inspection.

16. Motor Vehicle

a. All motor vehicles must be fit for the purpose of its use, commercial in nature, and have a valid driving license, vehicle insurance, and periodic maintenance records.

b. Heavy vehicles must be less than 10 years old, and passenger vehicles less than 5 years old.

c. All vehicles must have good tires, engine condition, working lights and horns, 3-way mirrors, visible registration number, good body condition, good and fail-safe brake systems, retractable 3-point seat belts, and ABS and airbags for passenger vehicles.

d. Drivers must be at least 25 years old with 5 years driving experience and good eyesight. They shall not drive for more than 12 hours.

e. Vehicles transporting hazardous goods must comply with all statutory obligations, including approval from the Chief Controller of Explosives and Transport Emergency (TREM) card and training.

f. Passenger transportation is only allowed in the passenger compartment. All passengers and the driver must wear seat belts.

g. Drivers must not use mobile phones or be under the influence of alcohol, drugs, or other narcotics while driving.

h. Drivers must not exceed the posted speed limit. The site speed limit for construction vehicles/cranes is 15 km/h.

i. Loads must not exceed the vehicle's carrying capacity.

j. Tractor trailers used for GPSR work shall not transport passengers unless approved by the site team.

k. Tractors with trailers/tankers must be driven at less than 20 KMPH or have speed governors, have 4-wheeled tankers with a capacity over 2000 Liters, and have either auto-disconnector or 360-degree rotation type connector pins.

17. Heavy Earth Moving Equipment (HEME)

a. All HEME supplied or used at the sites shall have mandatory safety attributes including rollover protection structure (ROPS), fall over protection structure (FOPS), automatic reverse alarm, fire extinguisher, functional wipers, heating and air conditioning vent (as applicable), and illumination lamp.

b. Operators must have a minimum 3 years of experience with HME or LME and be trained in safety procedures, including maintaining distance from overhead power lines and people.

c. Drivers must stop the engine and exit the cabin for any discussions.

d. Operators must wear all PPEs, avoid movement near trench/excavation edges, park the vehicle with all levers locked, and conduct daily and periodic inspections.

18. Concrete Vibrators (should conform Indian Standard-3558)

a. The vibrating unit shall be completely enclosed.

b. The belt transmitting power to the unit shall be adequately guarded.

c. Electrically operated compactor vibrators shall be totally enclosed and protected against overloads by suitable overload relays, and shall be effectively earthed.

d. Sufficient length of cable to the vibrator shall be ensured.

e. Needle load shall be firmly locked while inserting in the vibrator.

f. The inner core shall be adequately lubricated.

19. Concrete Mixers

a. All gears, chains and rollers of concrete mixers should be adequately guarded to prevent damage/danger.

b. Concrete mixer hoppers shall be protected by side railing to prevent workers from passing under them. Operators shall make sure before lowering the skip bucket that no personnel are present there.

c. Hopper hoist, wire rope and anchoring brake, and skip hoist clutch, must be checked for its condition and adjusted periodically. Nothing should be kept inside the motor enclosure.

d. The operator shall ensure the motor fan guard is secured firmly, wiring is properly connected and insulated and double earthing is done for electric mixers.

e. Waste RMC must be poured inside the batching plant area only. Cleaning of RMC shall be done within the dedicated area/ inside the batching plant area.

20. Access Control Security System

a. The contractor is responsible to provide valid identity cards to its personnel, as approved by GPSR's site team.

b. These cards should indicate basic details about the contractor's personnel and should only be issued when the contractor's personnel have attended all safety orientation/induction training and passed health screening.

21. Personnel Access, Vehicle and Material/Equipment Movement Control

a. All personnel, vehicles, and materials shall move in and out of the Site only through designated security checkpoints.

b. All personnel must register their entry and exit at the security checkpoint.

c. Materials/equipment moving in and out of the Site shall be accompanied by an approved gate pass.

d. All personnel must declare personal property (e.g., computers, tools, equipment, samples) at security checkpoints using the Material/Equipment Gate Pass before entering the Site.

e. Approval is required from authorized signatories before removing any undeclared personal property from the Site.

22. Education and Training

a. All contractor personnel are required to attend site safety orientation/induction training, prior to entering the site.

b. The following training and information must be provided during the orientation:
   i. Job rules, personal safety and conduct;
   ii. GPSR lifesaving safety rules, hazards reporting;
   iii. Reporting of injuries/pre-existing conditions; emergency procedures;
   iv. Working with chemicals;
   v. Firefighting;
   vi. Safety activities and program including disciplinary measure and incentives; and
   vii. JSA and PTW as outlined above.

c. Site and job-specific training shall be imparted as per specifications.

23. Appointment of Safety Resources, Officer/Supervisors and Safety Stewards.

a. The contractor shall appoint competent safety resources for the job

b. The contractor shall ensure availability of one safety supervisor in case of up to 20 deployed personnel, and an additional safety supervisor for personnel numbering between 20 and 50.

c. If there are more than 50 personnel, then a minimum of one safety engineer/officer is required to be deployed. For every additional 50 personnel, an additional safety supervisor shall be deployed.

d. The contractor should also deploy trained safety stewards – one for every 50 personnel – to ensure safety vigilance.

24. Supervisor’s Role in Worker's Orientation

The attitude of the workers towards accident prevention is directly defined by the attitude of the supervisor. As part of a new worker's orientation, the supervisor shall:

a. Ask about the worker's last job and describe the new job.

b. Show the worker around the work area and point out hazards.

c. Introduce the worker to others and describe basic rules.

d. Procure a history of the worker's previous training and provide understanding on JSA.

e. Give the worker a test run on tools and equipment.

f. Monitor the new worker's safety, health, and environmental performance.

g. Provide coaching where necessary.

h. Check back to see how the worker is progressing.

The contractor shall establish systems to check the competency of skilled workers and supervisors.

25. Pre-employment checks for site personnel

a. The contractor shall ensure that personnel (engineers/supervisor/skilled/semiskilled/unskilled tradesmen) are appointed based on their previous performance and following background checks.

b. The personnel must have the necessary licenses issued by governmental authorities or other competent bodies – electrician wireman, drivers (LMV/HMV), equipment operators, radiography, sand blasting or any other as per statutory requirements.

26. Pre-employment medical check up

a. The contractor shall carry out a pre-employment medical check-up for all personnel as per **GPSR's Occupational Health Guideline**
b. Certificates of fitness obtained after such check-ups must be provided to GPSR in the prescribed format prior to the deployment of personnel at site.
c. The contractor shall also organize periodical medical check-ups (half-yearly) for the following category of personnel:
    i. Drivers and equipment operators: Check for vision and hearing
    ii. Height workers: Check for vision, hearing, vertigo and acrophobia
    iii. Dangerous substances handlers: Effect of chemicals and substance.

27. Personal Protective Equipment (PPE) protocol

a. The contractor shall ensure availability and use of suitable quality PPEs for its personnel.

b. Safety helmets, safety shoes, high visibility jackets, and safety eye wear are mandatory.

c. Other PPEs, as indicated in the Work Permit and Job Safety Analysis, shall be used based on job requirements.

d. All PPEs used shall be in accordance with GPS PPE specifications document as SOP IS codes.

e. The contractor shall obtain prior GPS approval for any deviation in PPE standards.

28. Working over or near water bodies

a. The contractor must comply with GPSR's standard requirement on working over or near water bodies, including medical suitability, necessary statutory certification, use of life jackets, involving a skilled team, a rescue system in place in case a person falls in the water and use of insulated powered tools.

29. Pre-Startup Safety Reviews (PSSR)

a. PSSR is a process to validate whether:
   i. The construction and equipment installed are in accordance with design specifications
   ii. The risks associated with operation of the plant has been identified, studied and discussed
   iii. The preventive and corrective actions have been completed\
   iv. Any emergency actions which can be required have been understood, documented and that the plant/facility is complete in all safety aspects, before the start-up of the facility.

b. The contractor shall be responsible to re-work processes based on the findings of the PSSR.

31. Notice Boards/Caution Boards/Safety Awareness Boards

a. A safety board must be present, containing: manpower numbers; date of last incident; total number of injuries, incidents, and near misses; million man-hours injury-free; emergency contact details; current severity index; and number of unsafe acts and conditions.

b. The contractor shall display a site layout, indicating: project manager's and contractor's details; job details and dates; pathways; first-aid post; drinking water; change room; toilet; entry and exit points; and parking.

c. Background colour coding: blue for mandatory, red for prohibited, yellow for danger, green for safe condition.

d. Minimum 2 safety signages should be displayed at every 20-meter periphery area.

32. Safety Awareness Campaign/ Program

a. Contractors are encouraged to organize programs and campaigns promoting safety best practices

33. Medical/First Aid Treatment

a. The contractor shall maintain a first aid box and/or medical centre as per statutory requirement

b. The contractor shall have a tie-up with nearby good specialty hospitals to manage medical emergencies

c. A medical attendant shall administer first-aid care to injured personnel using established standards of practice, or as directed by the local consulting physician.

d. The contractor shall ensure injured personnel requiring treatment beyond the medical attendant's limits are transported to the nearest hospital.

e. If available, GPSR may, at its option, make its first-aid centre and other similar facilities at the site available for treatment of the contractor's personnel injured or ill while engaged in the performance of the scope of work.

34. Accident Investigation and Reporting

a. GPSR's policy requires that all incidents and environmental releases, regardless of severity, are reported immediately to the GPSR project manager and safety manager, first by verbal and then by a written "preliminary" First Incident Report (FIR)

b. Subsequently, a joint written investigation should be conducted by the incident investigation team, with an initial report issued in writing within 24 hours, even where no injuries occur.

c. The incident must be recorded by the GPSR team on the HSE application

d. In the event of the actual or potential severity of an incident, or repetitive unsafe behaviour, or increased frequency of such events, the project head will appoint a committee to investigate and complete a root cause analysis

e. All corrective actions shall be carried out, monitored by the project/construction department and initialled/dated on the original report.

35. Contractor inspections and audits

a. The contractors' supervisors are responsible for ensuring that regular inspections are made of work areas/activities

b. Project safety managers or their designates shall regularly audit project facilities and report deviations to GPSR.

c. This audit shall also involve the 'Safety Risk Field Audit' (SRFA) based on the severity index.

36. Contractor Safety Assurance Audit

a. Copies of the 'Weekly Safety Walkthrough' and inspection reports must be forwarded to the respective project manager as well as to the contractor's site management team for corrective actions

b. Contractor site management must take immediate corrective actions as required.

37. Periodic Safety Inspection

a. The contractor and site supervisors must conduct safety inspections daily, and provide updates regularly, as per GPSR's inspection and monitoring SOP.

b. These inspections must make note of; hazardous material, unsafe acts/conditions observations and corrections, equipment checks, rigging, electrical checks, scaffolding status, ladder inspections, etc.

38. Safe-work practice assessments

a. The project safety managers shall conduct periodic assessments of compliance with site safety practices and procedures to ensure that they are effective and being followed.

39. Fire prevention and protection

a. GPSR's fire prevention policies require prevention and mitigation training on fire emergencies, designating segregated storage areas for flammable and combustible material, safety practices for welding and burning operations, high risk welding operations, etc.

b. The contractor is mandated to comply with these policies and conforms to IS-2190

40. Hazard Communication

a. The contractor shall ensure correct hazard communication practices are followed by the working team

b. These include maintenance of hazardous chemicals and their safety data sheets, precautions while handling and storing gas cylinders, performing hot work, entering and working in confined spaces, working in an excavated pit, identifying and correct usage of quality PPEs, high noise and vibration.

c. This communication should be carried out through verbal, written, JSA, work permits, training, tool box talks, symbol display, practical display and mock demonstration.

41. Housekeeping Guidelines

a. Contractors are responsible for maintaining high standards of housekeeping. These include:
   i. Clearly defined walkways and stairways
   ii. Clean unobstructed pathways free of tripping hazards
   iii. Prohibition of storage of materials and equipment in aisles or under stairways
   iv. Posting signs or barricades to warn of tripping or slipping hazards
   v. Maintaining adequate lighting and well-marked first aid/assembly points
   vi. Not allowing loose materials at height (placing them in cage/box)

b. Storage Areas: The contractor shall clearly identify the purpose of the area and define boundaries by barricades, provide adequate size as per provision and discourage excess material and equipment. All material shall be stored in a neat, safe and orderly fashion with proper labelling.

c. Trash removal plan: The contractor shall ensure trash is removed, correctly collected and dumped at the designated disposal area on a daily basis. Collected trash and bins shall be cleared once a week. No trash shall be thrown from height and shall be handled through a chute. As a best practice, GPSR appreciates the concept of "generator should be the cleaner". Segregation of waste shall be done.

d. Rest rooms and drinking water:
   i. The contractor shall ensure clean toilets and drinking water are available for their personnel
   ii. There must be separate facilities for different genders
   iii. The workmen restroom must be of a 300 sq. ft area for 100 workmen.

iv. If GPSR has developed such facilities at the site, it may make them available to the contractor's personnel.

42. Governance System

a. The contractor shall participate in the GPSR site-level safety committee.

b. The committee, chaired by the GPS site in-charge, includes management and workmen.

c. The committee shall review safety performance weekly, including leading and lagging indicators.

d. The committee shall submit safety-related MIS to the concerned site in-charge.

43. Emergency Response Plan

a. The contractor shall prepare an emergency plan in line with foreseen emergency situations and communicate it to staff as part of training.

b. Preparedness should be checked for effectiveness by conducting mock drills.

c. The emergency plan shall be approved by the Site In-charge.

d. An emergency rescue plan with an adequately trained team must be in place for work at height, confined spaces, hot jobs, etc.

e. The contractor shall submit the emergency rescue plan to GPSR for inclusion in the site-specific emergency response procedure.

44. General

a. The contractor shall provide proper illumination at work locations, as per statutory requirements and/or the requirements of GPSR.

b. The contractor shall coordinate and procure requisite approvals from the statutory authorities prior to any blasting work that is to be executed at the site.

c. Labour camp (if applicable): The contractor shall ensure that its personnel follow the requirements pertaining to health and sanitation, including requirement of maintaining hygiene on site. The contractor shall be responsible to follow all the statutory EHS requirements inside the labour camp.

45. Reward for Safety Compliances.

a. There will be a reward system for reporting and correcting UA/UC/NM cases, achievement of targeted severity index and excellent contribution in maintaining safety and housekeeping.

46. Gas Cylinder Storage (should conform to IS-4379 & IS-8198)

a. The floor of the gas cylinder storage area shall be level and free from undulations.

b. Gas cylinders shall be kept away from sources of heat.

c. Cylinders shall be stored in a well-protected, well-ventilated, dry location, away from highly combustible materials.

d. Cylinders shall be stored in assigned places away from stairs or passageways where they won't be knocked over or damaged.

e. Cylinders shall not be kept in unventilated enclosures.

f. Cylinders containing oxygen and combustible gases shall not be stored in the same room.

g. All gas cylinders shall be stored upright and chained to prevent falling.

h. All cylinders shall be protected against excessive temperature rises and inclement weather.

I. Warm (not boiling) water shall be used to thaw ice in cylinder valve outlets.

j. Cylinders containing oxygen shall be stored 20 feet away from fuel-gas cylinders or separated by a fire-resistant wall.

k. The storage area construction shall be of non-combustible material.

l. Full cylinders shall be used in rotation as received from the supplier.

m. Empty cylinders shall be kept separate from full ones and marked distinctively.

n. Full and empty cylinders shall have their valves closed.

o. Valve protection caps shall always be in place, except when cylinders are in use.

p. Conspicuous signs shall be posted forbidding smoking, open lights, or other open flames.

47. Oil Storage:

a. Storage quantity shall not exceed 900 litres by any means

b. Isolated area with shed and PCC floor with soak pit shall be prepared for storage.

c. Adequate capacity and quantity of foam type fire extinguishers shall be kept outside the storage area.

d. Closed containers shall be used for storage.

e. Dip trays shall be provided for below the storage containers.

f. Spill kit shall be kept to control spillage.

g. No Hot work shall be performed near the storage area at any point.

47. Safety Park:

a. A safety park like set-up shall be made by the contractor to create safety awareness among workers.

b. GPSR's HSE team will assist in this regard.

48. HSE Plan:

Contractors must submit project HSE plans to GPSR.

49. Penalty system for HSE violations

Failure to comply with the aforementioned HSE requirements may result in penalties. These penalties will be determined by the central HSE team of GPS Renewables based on regular site monitoring, data analysis, and geo-tagged photographic evidence recorded in the HSE Application managed by GPS Renewables. The severity and frequency of violations will be considered when imposing fines, which will be deducted from the contractor's dues.

Sl.no	Violation of HSE norms	Penalty Amount
1	Not using PPE (Helmet, Shoes, Goggles, Gloves, Full body harness, Face shield, Boiler suit, etc.)	Rs.500/- per day/ item / Person.
2	Working without Work Permit/Clearance	Rs 1,00,000/- per occasion
3	Execution of work without deployment of requisite field engineer / supervisor at work spot	Rs. 5,000/- per violation per day
4	Unsafe electrical practices (not installing ELCB, using poor joints of cables, using naked wire without top plug into socket, laying wire/cables on the roads, electrical jobs by incompetent people, etc.)	Rs.10,000/- per item per day.
5	Working at height without full body harness, using non-standard/ rejected scaffolding and not arranging fall protection arrangements as required, like hand- rails, life-lines, Safety Nets etc.	Rs.50,000/- per case per day.
6	Unsafe handling of compressed gas cylinders (No trolley, jubilee clips, double gauge regulator, and not keeping cylinders vertical during storage/handling, not using safety cap of cylinder).	Rs.500/- per item per day.

7	Use of domestic LPG for cutting purpose / not using flash back arresters on both the hoses/tubes on both ends.	Rs-3,000/- per item per Occasion
8	No fencing/barricading of excavated areas / trenches.	Rs.3000/- per item per Occasion
9	Not providing shoring/strutting/proper slope and not keeping the excavated earth at least 1.5M away from excavated area	Rs.5,000/- per occasion.
10	Non display of scaffold tags, caution boards, list of hospitals, emergency services available at work locations.	Rs.1000/- per occasion per day
11	Traffic rules violations like over speeding of vehicles, rash driving, talking on mobile phones during vehicle driving, wrong parking, not using seat belts, vehicles not fitted with reverse horn / warning alarms / flicker lamps during foggy weather.	Rs. 2000/- per occasion
12	Utilization of hydra/back-hoe loader for material shifting or any other unauthorized /unsafe lifting works.	Rs25,000/- per occasion
13	Failure to maintain HSE records by Contractor Safety personnel, in line with approved HSE Plan/Procedures/Contract specifications.	Rs10,000/- per month.
14	Poor Housekeeping	Rs. 5000/- per occasion
15	Failure to report &follow-up accident (including Near Miss) reporting system within specific time-frame.	Rs. 20,000/- per occasion
16	Degradation of environment (not confining toxic spills, spilling oil/lubricants onto ground)	Rs10,000/- per occasion
17	Carrying out sand blasting instead of grit/shot blasting	Rs. 50,000/- per day
18	Not medically examining the workers before allowing them to work at height / to work in confined space / to work in shot-blasting / to work for painting / to work in bitumen or asphalt works, not providing ear muffs while allowing them to work in noise polluted areas, made them to work in air polluted areas without respiratory protective devices, etc.	Rs 5000/- per occasion per worker
19	Violation of any other safety condition as per job work permit and HSE conditions of contract (e.g. using crowbar on cable trenches, improper welding booth, not keeping fire extinguisher ready at hot work site, unsafe rigging practices, non-availability	Rs 5000/- per occasion

	of First-Aid box at site, not using hood with respiratory devices by blaster for shot//grit blasting, etc.)	
20	Failure to deploy adequately qualified and competent Safety Officers.	Rs. 10,000/- per day per Officer
21	Any Medical Treatment Case- Physical injury like Cut Injuries which required stitching of skin.	Minimum Rs 2,00,000/-per injury
22	Any Reportable Physical Injury (Lost Time Injury) (0.1% of awarded contract price), {Accident injured a man & keep him away from work for 48 hours or more (excluding the day of accident)}	Minimum Rs 500000/- per injury
23	Any Fatal Accident (0.5% of awarded contract price + ESCI compensation)	Minimum Rs 800000/- per injury
23	Any violation not covered above	To be decided by GPS / GPS's Client
24	Unauthorized dumping of scrap	Rs 5000/- per occasion